

P5 — Missing-data sensitivity

Week 7 · applies L6 (Missing data) · optional and ungraded — no separate deadline, nothing to submit; M1 is assembled from these five increments Keep in your group GitHub repo: a missingness audit + one justified missing-data analysis compared with complete-case.

The question this week

Which reductions in your sample are actually missing data, how much genuine missingness is there, and does handling it change your main estimate?

The trap this increment exists to defuse: **a shrinking N is not evidence of a missing-data problem**. Cohorts shrink for several reasons and only one of them is missingness.

Step 1 — Classify every reduction before naming any of it “missing”

Go back to your P1 funnel and give every downward step a type. Do this **first**, before you quantify anything.

Type	What it is	Can a missing-data method fix it?
DESIGN	The study deliberately measured this variable in a subsample, and supplies a weight for it	No — and it does not need fixing. Use the right weight (P4).
ELIGIBILITY	A definitional restriction: age, calendar window, geography	No — it states who the study is about
TARGET POPULATION	A disease or exposure definition: people who do not have the condition	No — they are not who the paper is about
OUTCOME ASCERTAINMENT	Failure to link or follow up for the outcome — the record was never in a position to carry a value	No — count it, report it separately, and do not impute it
MISSINGNESS	Someone who <i>should</i> have a value and does not	Yes — this is the only row a missing-data method addresses

Designed subsampling is not missing data. If your survey supplies a subsample weight, absence outside that subsample is a design feature with a documented fix, not an uncorrectable selection bias. Inference remains conditional on the weighting assumptions — say so — but do not call it MNAR.

Step 2 — Name the problem you will analyse, and scope the rest

Your sensitivity analysis runs on the MISSINGNESS rows only: records that should have carried a value on a variable your locked model uses and do not. That is exactly what M0 criterion 6 required you to point at, and it is the only loss P5 addresses with the methods this course teaches. It must arise from your locked scientific analysis — the variables your main model already uses, not variables added until a problem appears.

The other rows stay in your write-up. DESIGN is answered by the weight you chose at P4; inference stays conditional on the weighting assumptions — say so, and do not call it MNAR. ELIGIBILITY and TARGET POPULATION state who the paper is about; report them so nobody reads them as attrition. OUTCOME ASCERTAINMENT is failure to link or follow up — the record was never in a position to carry an outcome value, so count it, report it separately, and do not impute it. A record that was followed and still has no outcome value is missingness, and belongs here in Step 2. Where any of these limits who your estimate applies to, write the one sentence that states the population your conclusion is about.

None of this says every loss above is unanalysable. Design is already answered, by the P4 weight; eligibility and target population state who the study is about, so there is nothing there to analyse. Selection beyond the design weight, and failure to link or follow up, do have methods — selection models and censoring weights among them — but they are not the methods this course teaches, so at P5 you classify, report and scope those losses rather than analyse them.

Adding covariates to manufacture missingness changes your adjustment set and answers a different question. That is not a missing-data analysis.

What to do

1. **Classify** every funnel step by type (Step 1). One table, five categories.
2. **Identify** the genuine missingness arising from your locked model (Step 2).
3. **Quantify** it: which variables, how many observations each, overlap between them, and the loss both unweighted and weighted.
4. **State your estimand** before comparing anything — complete-case and any alternative can silently target different populations.
5. **Choose a method** and state the assumptions that would make it informative.
6. **Measure whether it matters.** Compare against your P4 design-aware estimate and say whether the difference is *practically* meaningful, not merely whether one is “significant”.

Deliverable

- The **typed funnel table** — this becomes your M1 missingness-audit slide.
- A quantification of the genuine missingness in your locked model’s variables.
- Your complete-case estimate against one justified alternative, with the estimand and assumptions stated, and two or three sentences on whether it changed anything.
- One sentence naming the population your conclusion is about, given every restriction in the funnel. Not a caveat added if there is room — part of the deliverable.

Worked example — MASLD/NHANES demonstration

- **Classified:** after eligibility, the two largest reductions are DESIGN (the fasting subsample, handled by WTSAF2YR) and TARGET POPULATION (people without steatosis). Neither is missing data. Genuine missingness across the whole funnel is **792 item-nonresponse losses** and, separately, **2 outcome-ascertainment losses** — different problems, not added together.
- **Identified:** within the locked domain — the 6,048 rows P5 analyses — exactly **one** Model-1 covariate carries any missingness: cancer status, **109 observations, 1.8% unweighted, 1.18% weighted.**
- **Measured:** complete-case **2.97**, both extreme assignments of the missing values **2.98 / 2.99**, multiple imputation **2.98**. Everything inside a 0.02 band.
- **Concluded:** this missing-data problem mattered little **after we measured it.** The analysis was necessary because we did not know that in advance.

What good looks like

- **Exceptional:** classifies every funnel step by type before quantifying anything; identifies missingness arising from the locked model rather than manufactured; states the estimand and the assumptions; measures the effect and reports it honestly — including when the honest answer is “no material change” — and states in one sentence the population the conclusion is about.
- **Adequate:** quantifies the missingness in the model’s variables, runs complete-case against one alternative, interprets sensibly.
- **Weak:** treats total cohort shrinkage as the missing-data problem; calls designed subsampling MNAR; adds variables until a missing-data problem appears; reports “<10% missing, complete-case is fine” with no analysis; builds the increment on a loss P5 does not address with the methods this course teaches — designed subsampling, eligibility, the disease definition, or a linkage failure.

Be ready to defend (M3)

“Point at one step in your funnel and tell me whether it is design, eligibility, target population, outcome ascertainment, or missingness — and how you know.”

AI co-pilot note

A co-pilot will typically report the small per-variable missingness, run a quick `mice()` with default settings, and declare victory — never noticing the large *selection* into the analytic sample. Justify the imputation model and follow the selection upstream yourself.

Lab → project handoff

You are applying **L6 (Missing data)** — examining the missingness pattern and using multiple imputation with design-adjusted regression (EpiMethods `missingdataE`). Choose the imputation model and the number of imputations to fit *your* problem, not the lab’s fast defaults.